

Geometry Test 2 — Think First, Then Compute

Mini-Review (use these!)

Distance (2D): $d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
Section (internal) in ratio $m:n$:
 $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
Fraction $\frac{a}{b}$ from A toward B :
 $P = A + \frac{a}{b}(B - A)$

Angles: straight 180° ; right 90° ; vertical $\cong$;
 linear pair $\Rightarrow$ supplementary; bisector splits equally.
Perimeter/Area:
 Rect: $P = 2(\ell + w)$, $A = \ell w$; Square: $P = 4s$, $A = s^2$;
 Tri: $A = \frac{1}{2}bh$; Circle: $C = 2\pi r$, $A = \pi r^2$.

1. **(Fractional point on a segment)** $A(-3, 5)$, $B(9, -7)$. Find the point P that is $\frac{3}{5}$ of the way from A to B (starting at A).

2. **(m:n divider)** $A(2, -1)$, $B(8, 5)$. Point P divides $\overline{AB}$ in the ratio $2 : 3$ (closer to A). Find P .

3. **(Midpoint & distance)** For $C(-6, 4)$ and $D(10, -2)$, find (i) the midpoint M , (ii) the distance CD .

4. **(Find the missing endpoint)** P is the midpoint of $\overline{AB}$. If $A(7, -3)$ and $P(2, 5)$, find B .

5. (**Circle from diameter**) The endpoints of a diameter are $E(-4, 2)$ and $F(6, 8)$. Find the circle's center, radius, circumference, and area.

6. (**Opposite vertices of a square**) Opposite vertices of a square are $S_1(1, 1)$ and $S_2(5, 5)$. Find the side length, perimeter, and area of the square.

7. (**Complete the rectangle**) $A(1, 2)$, $B(7, 2)$, and $C(7, -1)$ are three consecutive vertices of a rectangle (in order). Find D . Then find the rectangle's area and perimeter.

8. (**Right/Isosceles?**) Triangle T has vertices $P(0, 0)$, $Q(4, 3)$, and $R(-3, 4)$. Classify T as right/isosceles/neither (justify using distance), and find its perimeter.

9. (**Complementary angles**) Two complementary angles have measures x° and $(x + 18)^\circ$. Find both angles.

10. (**Supplementary linear pair**) Angles forming a straight line measure $(3x + 15)^\circ$ and

$(5x - 3)^\circ$. Find x and both angles.

11. (**Vertical angles**) Vertical angles have measures $(2y + 14)^\circ$ and $(5y - 13)^\circ$. Find y and the angle measures.

12. (**Angle bisector**) Ray $\overrightarrow{BD}$ bisects $\angle ABC$. If $\angle ABD = (x + 20)^\circ$ and $\angle DBC = (2x - 10)^\circ$, find x and $\angle ABC$.

13. (**Segment bisector idea**) Point $M(4, -1)$ is the midpoint of $\overline{XY}$. If $X(-2, 3)$, find Y .

14. (**Triangle area from coordinates**) $A(0, 0)$, $B(8, 0)$, $C(5, 6)$. Find the area of $\triangle ABC$ and the length of side AC .

15. (**Rectangle via distance**) $R(2, 1)$, $S(10, 1)$, $T(10, 7)$, $U(2, 7)$. Compute both diagonals

to verify they are equal, then give area and perimeter.

16. **(Find the ratio from a point)** $A(-2, 4)$ and $B(13, -1)$. Point $P(4, 2)$ lies on $\overline{AB}$. Determine the ratio $AP : PB$ (as $m : n$ with positive integers).

17. **(Fractional locator with parameters)** $A(a, b)$ and $B(c, d)$. Write a formula for the coordinates of P that is $\frac{2}{5}$ of the way from A to B . Then apply it to $A(-1, 4)$ and $B(9, -6)$.

18. **(Right angle check)** Show that the angle at Q is right for $P(-1, 2)$, $Q(3, 0)$, $R(7, -2)$ by using slopes or distances.

19. **(Perimeter with a diagonal)** A rectangle has perimeter 38 and diagonal length 13. If the sides are integers, find the dimensions (length and width).

20. (**Circle from center to side length**) A square is inscribed in a circle of radius $r = 5$. Find the square's side length and area.
